Status	Finished
Started	Saturday, 10 October 2026, 10:42 PM
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Duration	4 mins 22 secs
Grade	**13.00** out of 15.00 (**86.67**%)

Question **1**
Complete
Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

- a. Extraction
- b. Selection
- c. Distribution
- d. Insertion

The correct answer is:
Insertion

Question **2**
Complete
Mark 1.00 out of 1.00

What is a characteristic of the merge sort algorithm?

- a. It is less efficient than bubble sort
- b. It does not require recursion
- c. It is based on the divide-and-conquer approach
- d. It sorts data using a single pass

The correct answer is: It is based on the divide-and-conquer approach

Question **3**
Complete
Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It requires external libraries
- c. It is less efficient than custom sorting algorithms
- d. It sorts data out of the box efficiently

The correct answer is: It sorts data out of the box efficiently

Question **4**
Complete
Mark 0.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 40,70,80,90,120,150,200,300,400,470
- b. 80,150,200,470,40,90,300,400,70,120
- c. 40,80,90,150,200,300,400,470,70,120
- d. 200,470,80,150,40,90,300,400,70,120

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Linear Search
- b. Merge Sort
- c. Binary Search
- d. Bubble Sort

The correct answer is: Bubble Sort

Question **6**
Complete
Mark 1.00 out of 1.00

What is Bubble Sort known for?

- a. Sorting data in a non-sequential manner
- b. Using the divide-and-conquer approach
- c. Being the most efficient sorting algorithm
- d. Bubbling up the largest element to its correct position with each pass

The correct answer is: Bubbling up the largest element to its correct position with each pass

Question **7**
Complete
Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Complexity
- c. Merging
- d. Sorting

The correct answer is:
Complexity

Question **8**
Complete
Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Quick Sort
- b. Merge Sort
- c. Bubble Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **9**
Complete
Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of all keys
- b. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- c. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- d. A sorting algorithm is stable if it preserves the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **10**
Complete
Mark 1.00 out of 1.00

What is a key disadvantage of Bubble Sort compared to more advanced algorithms like Merge Sort?

- a. Bubble Sort is less efficient for large lists
- b. Bubble Sort is difficult to implement
- c. Bubble Sort does not guarantee sorted order
- d. Bubble Sort cannot handle duplicate elements

The correct answer is: Bubble Sort is less efficient for large lists

Question **11**
Complete
Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It makes the data unsorted
- b. It complicates the selection of items
- c. It makes it easier to select items based on their relationship to the rest of the items
- d. It slows down the process

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **12**
Complete
Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

- a. order()
- b. arrange()
- c. sorted()
- d. sort()

The correct answer is: sorted()

Question **13**
Complete
Mark 1.00 out of 1.00

What does the Bubble Sort algorithm primarily focus on during each pass?

- a. Sorting the entire list in one pass
- b. Bubbling up the largest element to its correct position
- c. Bubbling up the smallest element
- d. Dividing the list into halves

The correct answer is: Bubbling up the largest element to its correct position

Question **14**
Complete
Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Bubble Sort
- b. Quick Sort
- c. Insertion Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **15**
Complete
Mark 1.00 out of 1.00

Which of the following best describes the process of Merge Sort?

- a. It repeatedly finds the minimum element and moves it to the sorted part of the list
- b. It compares adjacent elements and swaps them if necessary
- c. It divides the list into two halves, sorts each half, and then merges them
- d. It builds a sorted array one element at a time

The correct answer is: It divides the list into two halves, sorts each half, and then merges them